BACHELOR OF COMPUTER APPLICATION

SEMESTER 1

FUNDAMENTALS OF MATHEMATICS

Unit 11

Maxima and Minima

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1. INTRODUCTION

In Unit 10, we dove deep into the differentiation of various functions, from trigonometric and inverse trigonometric functions to exponential, logarithmic, and hyperbolic functions. You've mastered how to handle the derivatives of these essential functions, setting a solid foundation in calculus that connects beautifully with real-world applications. This knowledge is not just about solving equations; it's about understanding change, movement, and growth in any scenario, whether it be in physics, engineering, or finance.

Now, welcome to Unit 11, where we'll explore the practical and fascinating world of Applications of Derivatives. In this unit, you'll learn how these mathematical tools are not just academic exercises but are powerful techniques used to analyse real-world phenomena. From understanding the dynamics of a speeding car to predicting economic changes, derivatives are key to solving complex problems in science and economics. Expect to delve into rate changes, optimization problems, and even how derivatives help in constructing tangents and normals to curves.

To make the most of this unit, approach each concept with curiosity about its application in real life. We'll start with basic principles, then move to more complex applications like motion analysis and economic models. Engage with the examples provided—they're designed to bridge the gap between theory and practical application. By the end of this unit, you should be able to not only solve mathematical problems but also apply these concepts to understand and predict natural and economic phenomena effectively. Embrace the challenge, and let's see how far these derivatives can take us!

1.1. Learning Objectives

At the end of this unit, you will be able to:

- *Identify and recall the definitions of key derivative applications such as rate of change, optimization, and tangents.*
- *Explain the concepts behind velocity and acceleration as applications of first and second derivatives.*
- *Calculate the derivative of given functions to find the slope of tangent lines at specified points.*
- *Differentiate between problems that require the use of the product rule versus the quotient rule in derivative applications.*

2. APPLICATIONS OF DERIVATIVES

Derivatives have a wide range of applications in mathematics and other fields. They are used to analyse the rate of change of quantities, determine the behavior of functions, find tangents and normals to curves, approximate values, and identify maximum and minimum points. Below, we explore these applications in detail.

2.1. Rate of Change of Bodies

One of the primary applications of derivatives is to measure how a quantity changes with respect to another. This concept is fundamental in physics and other sciences.

2.1.1. Velocity and Acceleration

In physics, the position of a moving object is often given as a function of time, $s(t)$. The derivative of this function with respect to time gives the velocity, $v(t)$ of the object:

$$v(t) = \frac{d}{dt}s(t)$$

The derivative of the velocity function gives the acceleration, $a(t)$ of the object:

$$a(t) = \frac{d}{dt}v(t) = \frac{d^2}{dt^2}s(t)$$

Example:

If the position of a car is given by $s(t) = 5t^2 + 2t$ where $s$is in meters and t is in seconds, then:

The velocity is:

$$v(t) = \frac{d}{dt}(5t^2 + 2t) = 10t + 2$$

The acceleration is:

$$a(t) = \frac{d}{dt}(10t + 2) = 10m\ /\ s^2$$

2.2. Increasing/Decreasing Functions

Derivatives are used to determine whether a function is increasing or decreasing in a given interval.

- A function $f(x)$ is increasing on an interval if $f'(x) > 0$ for all x in that interval.
- A function $f(x)$ is increasing on an interval if $f'(x) < 0$ for all x in that interval.

Example:

Consider the function $f(x) = x^3 - 3x^2 + 2$. o determine where the function is increasing or decreasing:

1. Find the derivative:

$$f'(x) = 3x^2 - 6x$$

2. Set the derivative equal to zero to find critical points:

$$3x^2 - 6x = 0$$
$$x(x-2) = 0$$
$$x = 0,2$$

3. Determine the sign of $f'(x)$ in each interval defined by the critical points (eg., $(-\infty, 0), (0,2), an\, d(2, \infty)$).
   a. For $x \in (-\infty, 0), f'(x) > 0$ (increasing)
   b. For $x \in (0,2), f'(x) < 0$ (decreasing)
   c. For $x \in (2, \infty), f'(x) > 0$ (increasing)

2.3. Tangents and Normals

Derivatives are used to find the equations of tangent and normal lines to a curve at a given point.

2.3.1. Tangent Line

The slope of the tangent line to the curve $y = f(x)$ at the point $\big(a, f(a)\big)$ is given by $f'(a)$. The equation of the tangent line is;

$$y - f(a) = f'(a)(x - a)$$

2.3.2. Normal Line

The normal line is perpendicular to the tangent line. The slope of the normal line is the negative reciprocal of the slope of the tangent line:

$$slope\ of\ normal\ line\ = -\frac{1}{f'(a)}$$

The equation of the normal line is:

$$y - f(a) = -\frac{1}{f'(a)}(x - a)$$

Example:

For the function $f(x) = x^2$ at the point (1,1):

1. Find the derivative:

$$f'(x) = 2x$$

2. The slope of the tangent line at $x = 1$ is $f'(1) = 2$
3. The equation of the tangent line is:

$$y - 1 = 2(x - 1)$$
$$y = -\frac{1}{2}x + \frac{3}{2}$$

4. The slope of the normal line at $x = 1$ is $-\frac{1}{2}$
5. The equation of the normal line is:

$$y - 1 = -\frac{1}{2}(x - 1)$$
$$y = -\frac{1}{2}x + \frac{3}{2}$$

2.4. Use of Derivatives in Approximation

Derivatives are used in approximation techniques, such as linear approximation and Taylor series.

2.4.1. Linear Approximation

The linear approximation of $f(x)$ near $x = a$ is given by:

$$f(x) \approx f(a) + f'(a)(x - a)$$

Example:

Approximate $\sqrt{4 \cdot 1}$:

Let $f(x) = \sqrt{x}, a = 4$

$$f'(x) = \frac{1}{2\sqrt{x}}$$

Using x = 4:

$$f'(\Delta) = \frac{1}{2\sqrt{4}} = \frac{1}{2 \times 2} = \frac{1}{4}$$

The linear approximation is:

$$f(4 \cdot 1) \approx \sqrt{4} + \frac{1}{4}(4.1 - 4) = 2 + \frac{0 \cdot 1}{4} = 2.025$$

2.5. Maxima and minima

Derivatives are used to find the maximum and minimum values of functions.

2.5.1. First Derivative Test

The first derivative test is used to determine whether a critical point is a local maximum, minimum, or neither.

1. Find the critical points by setting $f'(x) = 0$.
2. Determine the sign of $f'(x)$ before and after each critical point.
    a. If $f'(x)$ changes from positive to negative, there is a local maximum.
    b. If $f'(x)$ changes from negative to positive, there is a local minimum.
    c. If $f'(x)$ does not change sign, the critical point is neither a maximum nor a minimum.

2.5.2. Second Derivative Test

The second derivative test provides a more precise method to determine the nature of critical points.

1. Find the second derivative $f''(x)$
2. Evaluate $f''(x)$ at the critical points
    a. If $f''(a) > 0, f(a)$ is a local minimum
    b. Id $f''(a) < 0, f(a)$ is local maximum
    c. If $f''(a) = 0$, the test is inconclusive

Example:

For the function $f(x) = x^3 - 3x^2 + 4$:

1. Find the first derivative:
$$f'(x) = 3x^2 - 6x$$
2. Set $f'(x) = 0$ to find critical points:
$$3x^2 - 6x = 0$$
$$3x(x-2) = 0$$
$$x = 0,2$$
3. Use the second derivative test:
$$f''(x) = 6x - 6$$
$f''(0) = -6$ (local maximum)

$f''(2) = 6$ (local minimum)

Derivatives have numerous applications in mathematics, physics, engineering, and other fields. They provide tools for analysing the rate of change of quantities, determining the behavior of functions, finding tangents and normals to curves, approximating values, and identifying maximum and minimum points. Mastery of these applications is essential for solving complex problems and understanding dynamic systems.

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3. ABSOLUTE MAXIMUM AND MINIMUM VALUES OF A FUNCTION IN A CLOSED INTERVAL

In calculus, finding the absolute maximum and minimum values of a function on a closed interval is a common problem with numerous applications in various fields, such as engineering, economics, and natural sciences. These values provide critical information about the range of a function within a specified domain.

Absolute Maximum: The highest value that a function attains on a given interval.

Absolute Minimum: The lowest value that a function attains on a given interval.

3.1. The Extreme Value Theorem

The Extreme Value Theorem guarantees that a continuous function on a closed interval will attain both an absolute maximum and an absolute minimum.

Theorem:

If f is continuous on a closed interval $[a, b]$, then f attains an absolute maximum value $f(c)$ and an absolute minimum values $f(d)$ at some numbers c and d in $[a, b]$.

3.2. Steps to Find Absolute Maximum and Minimum Values

To find the absolute maximum and minimum values of a continuous function f on a closed interval [a,b], follow these steps:

1. Find the critical points of f in (a,b) by solving $f'(x) = 0$ or where $f'(x)$ does not exist.
2. Evaluate the function at the critical points and at the endpoints a and b.
3. Compare the function values obtained in step 2. The largest value is the absolute maximum, and the smallest value is the absolute minimum.

Examples:

1. Polynomial Function

Find the absolute maximum and minimum values of the function $f(x) = x^3 - 3x^2 + 2$ on the interval $[-1,3]$.

Solution:

Find the Critical Points:

$$f'(x) = 3x^2 - 6x$$
$$3x^2 - 6x = 0$$
$$x(x-2) = 0$$
$$x = 0,2$$

Evaluate the function at the critical points and at the endpoints $x = -1$ and $x = 3$:

$$f(-1) = (-1)^3 - 3(-1)^2 + 2 = -1 - 3 + 2 = -2$$
$$f(0) = 0^3 - 3(0)^2 + 2 = 2$$
$$f(2) = 2^3 - 3(2)^2 + 2 = 8 - 12 + 2 = -2$$
$$f(3) = 3^3 - 3(3)^2 + 2 = 27 - 27 + 2 = 2$$

Compare the function values

$$f(-1) = -2$$
$$f(0) = 2$$
$$f(2) = -2$$
$$f(3) = 2$$

The absolute maximum value is 2 at $x = 0$ and $x = 3$, The absolute minimum value is -2 at $x = -1$ and $x = 2$.

2. Trigonometric Function

Find the absolute maximum and minimum values of the function $f(x) = sin\, x - cos\, x$ on the interval $[0,2\pi]$.

Solution:

Find the critical points:

$$f'(x) = cos\, x + sin\, x$$
$$cos\, x + sin\, x = 0$$
$$sin\, x = -\, cos\, x$$
$$tan\, x = -1$$
$$x = \frac{3\pi}{4}, \frac{7\pi}{4}$$

Evaluate the function at the critical points and at the endpoints $x = 0$ and $x = 2\pi$:

$$f(0) = sin(0) - cos(0) = 0 - 1 = -1$$
$$f\left(\frac{3\pi}{4}\right) = sin\left(\frac{3\pi}{4}\right) - cos\left(\frac{3\pi}{4}\right) = \frac{\sqrt{2}}{2} - \left(-\frac{\sqrt{2}}{2}\right) = \sqrt{2}$$

$$f\left(\frac{7\pi}{4}\right) = sin\left(\frac{7\pi}{4}\right) - cos\left(\frac{7\pi}{4}\right) = -\frac{\sqrt{z}}{2} - \frac{\sqrt{2}}{2} = -\sqrt{2}$$

$$f(2\pi) = sin(2\pi) - cos(2\pi) = 0 - 1 = -1$$

Compare the function values:

$$f(0) = -1$$

$$f\left(\frac{3\pi}{4}\right) = \sqrt{2}$$

$$f\left(\frac{7\pi}{4}\right) = -\sqrt{2}$$

$$f(2\pi) = -1$$

The absolute maximum value is $\sqrt{2}$ at $x = \frac{3\pi}{4}$. The absolute minimum value is $-\sqrt{2}$ at $x = \frac{7\pi}{4}$.

3. Real-Life Situation

A rectangular garden with a fixed perimeter of 40 meters is to be constructed. What dimensions will maximize the area of the garden?

Solution:

Set up the problem:

Let the length of the garden be l and the width be w. The perimeter is fixed at 40 meters, so:

$$2l + 2w = 40$$

$$l + w = 20$$

$$w = 20 - l$$

Express the area A in terms of one variable:

$$A = l \cdot w = l(20 - l) = 20l - l^2$$

Find the critical points:

$$\frac{dA}{dl} = 20 - 2l$$

$$20 - 2l = 0$$

$$l = 10$$

Evaluate the area at the critical point and at the endpoints:

Since the length and width must be positive, consider the endpoints $l = 0$ and $l = 20$:

$$A(10) = 20(10) - 10^2 = 200 - 100 = 100m^2$$

$$A(0) = 20(0) - 0^2 = 0m^2$$

$$A(20) = 20(20) - 20^2 = 400 - 400 = 0m^2$$

he absolute maximum area is 100 square meters, which occurs when the garden is a square with dimensions 10 meters by 10 meters.

Finding the absolute maximum and minimum values of a function on a closed interval is a crucial technique in calculus. This process involves finding critical points, evaluating the function at these points and the endpoints, and comparing the values. These methods are not only essential for understanding the behavior of mathematical functions but also have practical applications in fields such as engineering, economics, and natural sciences. Mastering these techniques allows one to solve a wide range of real-world problems effectively.

4. WORKING EXAMPLES

Derivatives are widely used to solve problems involving rates of change, behavior of functions, and optimization. Below are some simple problems that illustrate basic principles and understanding of derivatives, including real-life applications.

4.1. Rate of Change of Bodies

A car is traveling along a straight road. Its position at time t (in seconds) is given by $s(t) = 4t^2 + 3t$ (in meters). Find the car's velocity and acceleration at t=2 seconds.

Solution:

Find the velocity by differentiating the position function:

$$v(t) = \frac{ds}{dt} = \frac{d}{dt}(4t^2 + 3t) = 8t + 3$$

Find the velocity at $t = 2$:

$$v(2) = 8(2) + 3 = 16 + 3 = 19m / s$$

Find the acceleration by differentiating the velocity function:

$$a(t) = \frac{dv}{dt} = \frac{d}{dt}(8t + 3) = 8$$

The acceleration is constant at $8m / s^2$

So, the car's velocity at $t = 2$ seconds is $19m / s$, and its acceleration is $8m / s^2$.

4.2. Increasing/Decreasing Function

Determine the intervals on which the function $f(x) = x^3 - 3x^2 + 2$ is increasing or decreasing.

Solution:

Find the first derivative:

$$f'(x) = 3x^2 - 6x$$

Find the critical points by setting the first derivative equal to zero:

$$3x^2 - 6x = 0$$

$$x(x - 2) = 0$$

$$x = 0,2$$

Determine the sign of $f'(x)$ in the intervals defined by the critical points:

For $x \in (-\infty, 0)$; $f'(x) = 3x^2 - 6x > 0$ (Increasing)

For $x \in (0,2): f'(x) = 3x^2 - 6x < 0$ (Decreasing)

For $x \in (2, \infty): f'(x) = 3x^2 - 6x > 0$ (Increasing)

So, the function is increasing on $(-\infty, 0) \cup (2, \infty)$ and decreasing on $(0,2)$ ·

4.3. Tangents and Normals

Find the equations of the tangent and normal lines to the curve $y = x^2 + 1$ at the point $(1,2)$

Solution:

Find the first Derivative:

$$\frac{dy}{dx} = 2x$$

Find the slope of the tangent line at $x = 1$:

$$\left.\frac{dy}{dx}\right|_{x=1} = 2(1) = 2$$

The equation of the tangent line is:

$$y - 2 = 2(x - 1)$$

$$y = 2x$$

The slope of the normal line is the negative reciprocal of the slope of the tangent line:

$$slope\ of\ normal\ line\ = -\frac{1}{2}$$

The equation of the normal line is:

$$y - 2 = -\frac{1}{2}(x - 1)$$

$$y = -\frac{1}{2}x + \frac{5}{2}$$

So, the equation of the tangent line is $y = 2x$ and the equation of the normal line is $y = -\frac{1}{2}x + \frac{5}{2}$

4.4. Use of Derivatives in Approximation

Use linear approximation to estimate $\sqrt{4 \cdot 1}$

Solution:

Let $f(x) = \sqrt{x}$, and choose $a = 4$:

$$f(a) = \sqrt{4} = 2$$

$$f'(x) = \frac{1}{2\sqrt{x}}$$

$$f'(4) = \frac{1}{2 \cdot 2} = \frac{1}{4}$$

Use the linear approximation formula:

$$f(x) \approx f(a) + f'(a)(x - a)$$

Approximate $\sqrt{4 \cdot 1}$:

$$\sqrt{4 \cdot 1} \approx 2 + \frac{1}{4}(4 \cdot 1 - 4)$$

$$\sqrt{4 \cdot 1} \approx 2 + \frac{0 \cdot 1}{4}$$

$$\sqrt{4 \cdot 1} \approx 2 + 0.025 = 2.025$$

So, the linear approximation of $\sqrt{4 \cdot 1}$ is 2.025.

4.5. Maxima and Minima

Find the local maxima and minima of the function $f(x) = x^3 - 6x^2 + 9x + 1$ using the first and second derivative tests.

Solution:

Find the first derivative:

$$f'(x) = 3x^2 - 12x + 9$$

Find the critical points by setting the first derivative equal to zero:

$$3x^2 - 12x + 9 = 0$$

$$x^2 - 4x + 3 = 0$$

$$(x - 1)(x - 3) = 0$$

$$x = 1,3$$

Use the second derivative test to determine the nature of the critical points:

$$f''(x) = 6x - 12$$

At $x = 1$: $f''(1) = 6(1) - 12 = -6$ (Since $f''(1) < 0, x = 1$ is a local maximum.)

At $x = 3$: $f''(3) = 6(3) - 12 = 6$ (Since $f''(3) > 0, x = 3$ is a local minimum.)

Find the function value at these points:

$$f(1) = 1^3 - 6(1)^2 + 9(1) = 1 - 6 + 9 + 1 \ = \ 5$$

$$f(z) = 3^3 - 6(3)^2 + 9(3) + 1 = 27 - 54 + 27 + 1 = 1$$

So, the local maximum is at $x = 1$ with $f(1) = 5$, and the local minimum is at $x = 3$ with $f(3) = 1$.

4.6. Real-Life Situations

A company's profit $P(x)$ (in thousands of dollars) is modelled by the function $p(x) = -2x^2 + 24x - 50$, where x is the number of units sold (in thousands). Find the number of units that maximizes the profit and the maximum profit.

Solution:

Find the first derivative of the profit function:

$$P'(x) = -4x + 24$$

Find the critical points by setting the first derivative equal to zero:

$$-4x + 24 = 0$$

$$x = 6$$

Use the second derivative test to confirm the nature of the critical point:

$$P''(x) = -4$$

(Since $P''(6) < 0, x = 6$ is a local maximum.)

Find the maximum profit:

$$P(6) = -2(6)^2 + 24(6) - 50$$

$$P(6) = -72 + 144 - 50$$

$$P(6) = 22$$

So, the profit is maximized when 6,000 units are sold, and the maximum profit is $22,000.

A cylindrical tank with a radius of 5 meters is being filled with water at a rate of 3 cubic meters per minute. How fast is the height of the water increasing when the water is 10 meters deep?

Solution:

The volume of water in the tank is given by $V = \pi r^2 h$.

Since the radius r is constant at 5 meters, the volume depends on the height h:

$$V = \pi(5)^2 h$$

$$V = 25\pi h$$

Differentiate both sides with respect to time t:

$$\frac{dV}{dt} = 25\pi \frac{dh}{dt}$$

Given $\frac{dV}{dt} = 3$ cubic meters per minute, solve for $\frac{dh}{dt}$:

$$3 = 25\pi \frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{3}{25\pi}$$

So, the height of the water is increasing at a rate of $\frac{3}{25\pi}$ meters per minute when the water is 10 meters deep.

These simple problems illustrate the fundamental principles and applications of derivatives in various contexts, including physics, business, and everyday situations. By understanding how to apply derivatives to these problems, one can gain a deeper appreciation for the power and versatility of calculus in solving real-life challenges.

5. SUMMARY

As we wrap up our exploration of the Applications of Derivatives, it's essential to reflect on the profound impact these mathematical tools have on understanding and navigating the world around us. Derivatives aren't just symbols on a page; they are the very means by which we can decode the language of change—whether in motion, economics, or engineering.

Throughout this unit, we've learned how to wield derivatives to unveil the velocity and acceleration of moving objects—how they do not merely measure speed or the force of an object, but they allow us to predict future states, providing a glimpse into the unfolding story of moving bodies. Consider our example of a car whose position over time was modelled by a simple polynomial. By taking the derivative, we could determine its speed at any given moment, and a second derivative gave us its acceleration, offering insights into its dynamic behavior under various conditions.

Moreover, we've seen how derivatives help us ascertain where a function increases or decreases, leading us to discover local maxima and minima—critical points that tell us where a business might maximize profit or minimize cost. This capability is not just academic; it's a powerful tool for decision-making in business and economics, where understanding these turning points can mean the difference between profit and loss.

Tangents and normals, though geometric concepts, came alive as we applied derivatives to find exact lines that touch or intersect curves at specific points. These are not merely exercises in calculus; they are fundamental in creating simulations, designing objects, and even in the precise art of constructing visuals in video games and animations.

As you move forward, remember the practical exercises we tackled, like approximating values using linear approximations or employing the Taylor series for more complex functions. These are not just mathematical tricks but essential techniques in engineering and data science for handling real-world problems where exact solutions are unattainable or impractical.

Finally, embrace the challenge of applying these concepts to new situations. Whether it's in physics, measuring how quickly an object will reach a certain speed, or in business, determining the most cost-effective way to increase production, the applications of

derivatives are boundless. As you continue your mathematical journey, keep linking these abstract concepts to tangible examples and real-world applications. This connection not only deepens understanding but also sparks the kind of curiosity and inquiry that drives all scientific and mathematical exploration.

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6. SELF-ASSESSMENT QUESTIONS

1. The derivative of the position function $s(t)$ with respect to time t is called the _____ of the object.
2. A function is increasing on an interval if its derivative $f'(x)$ is _____ than zero over that interval.
3. The equation of a tangent line to the curve at a point can be given by $y - f(a) = f'(a)(x - a)$, where $f'(a)$ is the _____ at that point.
4. If $f''(x) > 0$ 0 at a critical point, the function has a local _____ at that point.
5. To approximate $\sqrt{4 \cdot 1}$ using linear approximation, you would evaluate at $a = 4$ because $\sqrt{4} = 2$ and $f'(4)$ = _____.
6. Which derivative provides the acceleration of an object when its position is given as a function of time?
   A. First derivative
   B. Second derivative
   C. Integral
   D. None of the above
7. The function $f(x) = x^3 - 3x^2 + 2$ is decreasing in the interval:
   A. $(-\infty,0)$
   B. $(0,2)$
   C. $(2,\infty)$
   D. $(0,3)$
8. Which of the following is true for a function's critical point to be a local maximum?
   A. $f'(x) > 0$ before and $f'(x) < 0$ after the point
   B. $f'(x) < 0$ before and $f'(x) > 0$ after the point
   C. $f'(x)$ does no change signs
   D. None of the above
9. The derivative of $f(x) = ln(x^2 + 1)$ is:
   A. $\frac{2x}{x^2+1}$
   B. $\frac{1}{x^2+1}$

C. $\frac{1}{2x(x^2+1)}$

D. $\frac{x}{x^2+1}$

10. What is the slope of the normal line to the curve at $x = a$ if $f'(a) = 2$?

A. -2

B. ½

C. -1/2

D. 2

7. TERMINAL QUESTIONS

1. Find the velocity of an object at $t = 3$ seconds if its position is given by $s(t) = 5t^2 + 2t$
2. Calculate the acceleration of the object whose position function is $s(t) = 5t^2 + 2t$ at $t = 4$ seconds.
3. Determine where the function $f(x) = x^3 - 3x^2 + 2$ is increasing and decreasing.
4. Write the equation of the tangent line to the curve $y = x^2 + 1$ at $x = 1$.
5. Write the equation of the normal line to the curve $y = x^2 + 1$ at $x = 1$.
6. Use linear approximation to estimate $\sqrt{4.1}$ using $f(x) = \sqrt{x}$ around $a = 4$.
7. Identify the local maxima and minima for the function $f(x) = x^3 - 6x^2 + 9x + 1$ using the first and second derivative tests.
8. Calculate the slope of the tangent line at $x = 1$ for the function $f(x) = \sin x - \cos x$.
9. Determine the absolute maximum and minimum values of the function $f(x) = x^3 - 3x^2 + 4$ on the interval $[-1,3]$.
10. Solve for x where the function $f(x) = \sin x - \cos x$ has critical points within the interval $[0,2\pi]$.

8. ANSWERS

Self-Assessment Questions

1. velocity
2. greater
3. slope
4. minimum
5. $\frac{1}{4}$
6. B) Second derivative
7. B) (0,2)
8. A) $f'(x) > 0$ before and $f'(x) < 0$ after the point.
9. A) $\frac{2x}{x^2+1}$
10. C) $-1/2$

Terminal Questions

1. Velocity at $t = 3$ seconds: $v(t) = 10t + 2$, so $v(3) = 10(3) + 2 = 32$ m/s. (Section 2.1.1)
2. Acceleration at $t = 4$ seconds: $a(t) = 10$ m/s^2. (Section 2.1.1)
3. Increasing/Decreasing Intervals: Increasing on $(-\infty, 0)$ and $(2, \infty)$, decreasing on (0,2). (Section 2.2)
4. Tangent Line Equation: $y - 2 = 2(x - 1)$, so $y = 2x$. (Section 2.3.1)
5. Normal Line Equation: $y - 2 = -\frac{1}{2}(x - 1)$, so $y = -\frac{1}{2}x + \frac{5}{2}$. (Section 2.3.2)
6. Linear Approximation of $\sqrt{4.1}$: $f(4.1) \approx 2 + \frac{1}{4}(0.1) = 2.025$. (Section 2.4.1)
7. Local Maxima and Minima: Local maxima at $x = 1$, local minima at $x = 3$. (Section 2.5.1 and 2.5.2)
8. Slope of Tangent Line: $f'(1) = \cos 1 + \sin 1$. (Section 2.3.1)
9. Absolute Maxima and Minima Values: Maximum at $x = 3$ and minimum at $x = -1$. (Section 2.5 .1 and 2.5.2)
10. Critical Points: Solve $\cos x + \sin x = 0$, leading to $x = \frac{3\pi}{4}, \frac{7\pi}{4}$. (Section 2.3.1)